KITABU QUEST

Form II

MATHEMATICS

Cover scene

students solving mathematics with charts and models with an elephant mascot

Elephant

- Clear lessons
- Worked examples
- Fun activities
- Real-life problems

GRADE

II

TZA TIE-BASIC aligned draft

Think - Explore - Achieve

Title Page

KITABU QUEST Form II Mathematics

Tanzania TIE-BASIC learner book

Mascot: elephant

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Tanzania learners.

How To Use This Book

This book helps Form II learners study Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use objects, drawings, number sentences, and worked examples before independent practice.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

As you finish each topic, return to the success criteria and correct one answer before moving on.

Indices Standard Form And Number Patterns: Lesson Opener

Learning focus: Indices Standard Form And Number Patterns

Country link: a Tanzania school, market, farming, building, transport, or data problem for Indices Standard Form And Number Patterns.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about indices standard form and number patterns

Inquiry questions:
- How can indices standard form and number patterns help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Indices Standard Form And Number Patterns: Learn

Indices Standard Form And Number Patterns teaches how numbers change in a regular way.

Core idea:
- Look at how each term changes to the next term.
- A pattern can add, subtract, multiply, divide, or combine steps.
- Write the rule before extending the pattern.

Key words:
- Indices Standard Form And Number Patterns
- Indices
- Standard
- Form
- Number
- Patterns
- method
- model

Indices Standard Form And Number Patterns: Worked Example

Worked example for Indices Standard Form And Number Patterns: Continue the pattern 6, 12, 18, 24, __, __.

1. Compare terms: 12 - 6 = 6, 18 - 12 = 6.
2. Rule: add 6 each time.
3. 24 + 6 = 30, 30 + 6 = 36.

Answer: 30, 36.

Indices Standard Form And Number Patterns: Vocabulary And Study Skill

Use these words and study moves before you practise Indices Standard Form And Number Patterns. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Indices Standard Form And Number Patterns
- Indices
- Standard
- Form
- Number
- Patterns
- method
- model

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Indices Standard Form And Number Patterns without a clear example, method, or evidence.

Indices Standard Form And Number Patterns: Guided Activity

Guided activity for Indices Standard Form And Number Patterns:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Indices Standard Form And Number Patterns: Practice And Correction

Practice for Indices Standard Form And Number Patterns:

Main outcome: solve and explain problems about indices standard form and number patterns.

1. Continue the pattern 5, 10, 15, 20, __, __.
2. Write the rule for the pattern.
3. Create a similar pattern that increases by 4 each time.

Answer check:
Answers: 25, 30. Rule: add 5 each time.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to indices standard form and number patterns and solve it.

Indices Standard Form And Number Patterns: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Indices Standard Form And Number Patterns.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Indices Standard Form And Number Patterns: Outcome Check

Outcome check for Indices Standard Form And Number Patterns: Solve and explain problems about indices standard form and number patterns.

1. Continue 7, 14, 21, 28, __, __.
2. Write the rule.
3. Create one new pattern with the same rule.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Direct And Inverse Proportion: Lesson Opener

Learning focus: Direct And Inverse Proportion

Country link: a Tanzania school, market, farming, building, transport, or data problem for Direct And Inverse Proportion.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about direct and inverse proportion

Inquiry questions:
- How can direct and inverse proportion help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Direct And Inverse Proportion: Learn

Direct And Inverse Proportion teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Direct And Inverse Proportion
- Direct
- Inverse
- Proportion
- method
- model
- unit
- answer

Direct And Inverse Proportion: Worked Example

Worked example for Direct And Inverse Proportion: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Direct And Inverse Proportion: Vocabulary And Study Skill

Use these words and study moves before you practise Direct And Inverse Proportion. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Direct And Inverse Proportion
- Direct
- Inverse
- Proportion
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Direct And Inverse Proportion without a clear example, method, or evidence.

Direct And Inverse Proportion: Guided Activity

Guided activity for Direct And Inverse Proportion:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Direct And Inverse Proportion: Practice And Correction

Practice for Direct And Inverse Proportion:

Main outcome: solve and explain problems about direct and inverse proportion.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to direct and inverse proportion and solve it.

Direct And Inverse Proportion: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Direct And Inverse Proportion.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Direct And Inverse Proportion: Outcome Check

Outcome check for Direct And Inverse Proportion: Solve and explain problems about direct and inverse proportion.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Linear Equations And Inequalities: Lesson Opener

Learning focus: Linear Equations And Inequalities

Country link: a Tanzania school, market, farming, building, transport, or data problem for Linear Equations And Inequalities.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about linear equations and inequalities

Inquiry questions:
- How can linear equations and inequalities help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Linear Equations And Inequalities: Learn

Linear Equations And Inequalities teaches how lines and angles describe shapes and directions.

Core idea:
- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:
- Linear Equations And Inequalities
- Linear
- Equations
- Inequalities
- method
- model
- unit
- answer

Linear Equations And Inequalities: Worked Example

Worked example for Linear Equations And Inequalities: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Linear Equations And Inequalities: Vocabulary And Study Skill

Use these words and study moves before you practise Linear Equations And Inequalities. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Linear Equations And Inequalities
- Linear
- Equations
- Inequalities
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Linear Equations And Inequalities without a clear example, method, or evidence.

Linear Equations And Inequalities: Guided Activity

Guided activity for Linear Equations And Inequalities:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Linear Equations And Inequalities: Practice And Correction

Practice for Linear Equations And Inequalities:

Main outcome: solve and explain problems about linear equations and inequalities.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:
Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to linear equations and inequalities and solve it.

Linear Equations And Inequalities: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Linear Equations And Inequalities.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Linear Equations And Inequalities: Outcome Check

Outcome check for Linear Equations And Inequalities: Solve and explain problems about linear equations and inequalities.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Graphs And Coordinate Geometry: Lesson Opener

Learning focus: Graphs And Coordinate Geometry

Country link: a Tanzania school, market, farming, building, transport, or data problem for Graphs And Coordinate Geometry.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about graphs and coordinate geometry

Inquiry questions:
- How can graphs and coordinate geometry help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Graphs And Coordinate Geometry: Learn

Graphs And Coordinate Geometry teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Graphs And Coordinate Geometry
- Graphs
- Coordinate
- Geometry
- method
- model
- unit
- answer

Graphs And Coordinate Geometry: Worked Example

Worked example for Graphs And Coordinate Geometry: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Tanzania classroom, market, garden, transport, or home example.

Graphs And Coordinate Geometry: Vocabulary And Study Skill

Use these words and study moves before you practise Graphs And Coordinate Geometry. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Graphs And Coordinate Geometry
- Graphs
- Coordinate
- Geometry
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Graphs And Coordinate Geometry without a clear example, method, or evidence.

Graphs And Coordinate Geometry: Guided Activity

Guided activity for Graphs And Coordinate Geometry:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Graphs And Coordinate Geometry: Practice And Correction

Practice for Graphs And Coordinate Geometry:

Main outcome: solve and explain problems about graphs and coordinate geometry.

1. Solve one graphs and coordinate geometry question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to graphs and coordinate geometry and solve it.

Graphs And Coordinate Geometry: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Graphs And Coordinate Geometry.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Graphs And Coordinate Geometry: Outcome Check

Outcome check for Graphs And Coordinate Geometry: Solve and explain problems about graphs and coordinate geometry.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Similarity Congruence And Geometric Proof: Lesson Opener

Learning focus: Similarity Congruence And Geometric Proof

Country link: a Tanzania school, market, farming, building, transport, or data problem for Similarity Congruence And Geometric Proof.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about similarity congruence and geometric proof

Inquiry questions:
- How can similarity congruence and geometric proof help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Similarity Congruence And Geometric Proof: Learn

Similarity Congruence And Geometric Proof teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Similarity Congruence And Geometric Proof
- Similarity
- Congruence
- Geometric
- Proof
- method
- model
- unit

Similarity Congruence And Geometric Proof: Worked Example

Worked example for Similarity Congruence And Geometric Proof: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Tanzania classroom, market, garden, transport, or home example.

Similarity Congruence And Geometric Proof: Vocabulary And Study Skill

Use these words and study moves before you practise Similarity Congruence And Geometric Proof. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Similarity Congruence And Geometric Proof
- Similarity
- Congruence
- Geometric
- Proof
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Similarity Congruence And Geometric Proof without a clear example, method, or evidence.

Similarity Congruence And Geometric Proof: Guided Activity

Guided activity for Similarity Congruence And Geometric Proof:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Similarity Congruence And Geometric Proof: Practice And Correction

Practice for Similarity Congruence And Geometric Proof:

Main outcome: solve and explain problems about similarity congruence and geometric proof.

1. Solve one similarity congruence and geometric proof question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to similarity congruence and geometric proof and solve it.

Similarity Congruence And Geometric Proof: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Similarity Congruence And Geometric Proof.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Similarity Congruence And Geometric Proof: Outcome Check

Outcome check for Similarity Congruence And Geometric Proof: Solve and explain problems about similarity congruence and geometric proof.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometry Basics: Lesson Opener

Learning focus: Trigonometry Basics

Country link: a Tanzania school, market, farming, building, transport, or data problem for Trigonometry Basics.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about trigonometry basics

Inquiry questions:
- How can trigonometry basics help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometry Basics: Learn

Trigonometry Basics teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Trigonometry Basics
- Trigonometry
- Basics
- method
- model
- unit
- answer
- check

Trigonometry Basics: Worked Example

Worked example for Trigonometry Basics: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Tanzania classroom, market, garden, transport, or home example.

Trigonometry Basics: Vocabulary And Study Skill

Use these words and study moves before you practise Trigonometry Basics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Trigonometry Basics
- Trigonometry
- Basics
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trigonometry Basics without a clear example, method, or evidence.

Trigonometry Basics: Guided Activity

Guided activity for Trigonometry Basics:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Trigonometry Basics: Practice And Correction

Practice for Trigonometry Basics:

Main outcome: solve and explain problems about trigonometry basics.

1. Solve one trigonometry basics question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to trigonometry basics and solve it.

Trigonometry Basics: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trigonometry Basics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometry Basics: Outcome Check

Outcome check for Trigonometry Basics: Solve and explain problems about trigonometry basics.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Probability And Statistics: Lesson Opener

Learning focus: Probability And Statistics

Country link: a Tanzania school, market, farming, building, transport, or data problem for Probability And Statistics.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about probability and statistics

Inquiry questions:
- How can probability and statistics help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Probability And Statistics: Learn

Probability And Statistics teaches how to collect, organize, read, and explain information.

Core idea:
- Data must answer a clear question.
- Tables, tally charts, pictographs, and bar charts make data easier to read.
- A good conclusion says what the data shows.

Key words:
- Probability And Statistics
- Probability
- Statistics
- method
- model
- unit
- answer
- check

Probability And Statistics: Worked Example

Worked example for Probability And Statistics: A class records favourite fruits: mango 8, banana 12, pineapple 6. Which fruit is most popular?

1. Read the table values.
2. Compare 8, 12, and 6.
3. 12 is the greatest number.

Answer: Banana is most popular.

Probability And Statistics: Vocabulary And Study Skill

Use these words and study moves before you practise Probability And Statistics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Probability And Statistics
- Probability
- Statistics
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Probability And Statistics without a clear example, method, or evidence.

Probability And Statistics: Guided Activity

Guided activity for Probability And Statistics:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Probability And Statistics: Practice And Correction

Practice for Probability And Statistics:

Main outcome: solve and explain problems about probability and statistics.

1. Ask 10 learners a simple question and make a tally table.
2. Draw a bar chart from the tally table.
3. Write one conclusion from the data.

Answer check:
Answer check: totals in the tally table and bar chart should match.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to probability and statistics and solve it.

Probability And Statistics: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Probability And Statistics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Probability And Statistics: Outcome Check

Outcome check for Probability And Statistics: Solve and explain problems about probability and statistics.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Form II Applied Revision: Lesson Opener

Learning focus: Form II Applied Revision

Country link: a Tanzania school, market, farming, building, transport, or data problem for Form II Applied Revision.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about form ii applied revision

Inquiry questions:
- How can form ii applied revision help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Form II Applied Revision: Learn

Form II Applied Revision teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Form II Applied Revision
- Form
- Applied
- Revision
- method
- model
- unit
- answer

Form II Applied Revision: Worked Example

Worked example for Form II Applied Revision: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Tanzania classroom, market, garden, transport, or home example.

Form II Applied Revision: Vocabulary And Study Skill

Use these words and study moves before you practise Form II Applied Revision. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Form II Applied Revision
- Form
- Applied
- Revision
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Form II Applied Revision without a clear example, method, or evidence.

Form II Applied Revision: Guided Activity

Guided activity for Form II Applied Revision:

1. Work with a partner and read the worked example.
2. Make a similar question using Tanzania classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Form II Applied Revision: Practice And Correction

Practice for Form II Applied Revision:

Main outcome: solve and explain problems about form ii applied revision.

1. Solve one form ii applied revision question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Tanzania home, school, shop, garden, road, game, or timetable example connected to form ii applied revision and solve it.

Form II Applied Revision: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Form II Applied Revision.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Form II Applied Revision: Outcome Check

Outcome check for Form II Applied Revision: Solve and explain problems about form ii applied revision.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Indices Standard Form And Number Patterns: Review Clinic 1

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Indices Standard Form And Number Patterns that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Direct And Inverse Proportion: Review Clinic 2

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Direct And Inverse Proportion that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Linear Equations And Inequalities: Review Clinic 3

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Linear Equations And Inequalities that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Graphs And Coordinate Geometry: Review Clinic 4

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Graphs And Coordinate Geometry that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Indices Standard Form And Number Patterns: Application Workshop 1

Application workshop 1 for Indices Standard Form And Number Patterns.

Grade move: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Direct And Inverse Proportion: Application Workshop 2

Application workshop 2 for Direct And Inverse Proportion.

Grade move: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Linear Equations And Inequalities: Application Workshop 3

Application workshop 3 for Linear Equations And Inequalities.

Grade move: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Graphs And Coordinate Geometry: Application Workshop 4

Application workshop 4 for Graphs And Coordinate Geometry.

Grade move: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Similarity Congruence And Geometric Proof: Application Workshop 5

Application workshop 5 for Similarity Congruence And Geometric Proof.

Grade move: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Trigonometry Basics: Application Workshop 6

Application workshop 6 for Trigonometry Basics.

Grade move: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Probability And Statistics: Application Workshop 7

Application workshop 7 for Probability And Statistics.

Grade move: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Form II Applied Revision: Application Workshop 8

Application workshop 8 for Form II Applied Revision.

Grade move: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Indices Standard Form And Number Patterns: Application Workshop 9

Application workshop 9 for Indices Standard Form And Number Patterns.

Grade move: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Direct And Inverse Proportion: Application Workshop 10

Application workshop 10 for Direct And Inverse Proportion.

Grade move: connect ideas across topics and explain causes, methods, or consequences; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Tanzania school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Glossary

Use this glossary to revise important words in Mathematics.

- Indices Standard Form And Number Patterns: Explain this term using an example from the book, your school, or your community.
- Direct And Inverse Proportion: Explain this term using an example from the book, your school, or your community.
- Linear Equations And Inequalities: Explain this term using an example from the book, your school, or your community.
- Graphs And Coordinate Geometry: Explain this term using an example from the book, your school, or your community.
- Similarity Congruence And Geometric Proof: Explain this term using an example from the book, your school, or your community.
- Trigonometry Basics: Explain this term using an example from the book, your school, or your community.
- Probability And Statistics: Explain this term using an example from the book, your school, or your community.
- Form II Applied Revision: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Mathematics that shows what you can now do independently.

Your project must include:
1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.